JACK CAIOLA

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EDUCATION

Bachelor of Science: Mechanical Engineering - Robotics and Autonomous Systems

Embry-Riddle Aeronautical University

Sep 2020 – May 2025 Daytona Beach, FL

Coursework included CATIA, SolidWorks, Ansys, Excel, and MATLAB.

Flight Training

Arrow Aviation

Training began Jun 2016 Danbury, CT

Flew Cessna Skyhawks and attended FAA ground school covering airspace, weather, and safety regulations.

SKILLS

Engineering Tools

CATIA | SolidWorks | Autodesk | Ansys | Excel | Embedded Systems | ArduPilot | Mission Planner | SAP | BabtechQ

Programming Languages

MATLAB | VBA | C++ | Rust | Python

Design and Prototyping

CAD | 3D Printing | Assembly Line Management | Systems Integration

Robotics

ROS 1 Noetic | ROS 2 Humble

Licenses

Class A Commercial Driver's License

LINKS

View this resume as a WebAssembly app:
https://www.jcaiola.com/resume/

View the source code:
github.com/jackcaiola/Resume

ABOUT ME

Mechanical engineering graduate with hands-on experience in mechanical design, robotics, UAV research, embedded systems, prototyping, and automotive manufacturing. Interested in advancing autonomous technologies through multidisciplinary research and practical engineering, with experience in chassis integration, product development, and workflow automation.

EXPERIENCE

Layout Designer - Chassis Integration

McNeilus Trucks & Manufacturing (Oshkosh Corporation)

Aug 2025 – Present

- Support chassis layout and systems integration design using SolidWorks.
- Designed new products in SolidWorks and supported product rollout.
- Built automations to consolidate chassis integration data and generate metrics for enforcing incoming chassis requirements across multiple OEMs.

Undergraduate Researcher

Embry-Riddle Aeronautical University

Jul 2020 – Dec 2025 Daytona Beach, FL

- Supported research into practical applications of thermal energy storage systems.
- Assisted in embedded system design for autonomous weather data acquisition.
- **ERAU GoAero Competition - Team Member**
  Jun–Oct 2025 · Embry-Riddle Flight Research Laboratory

  Designed C++ control scripts to process four time-of-flight sensor readings per second, model ground planes, assess landing viability, and calculate adaptive servo offsets for a four-legged autonomous drone. Integrated the scripts with a ROS 1 Noetic node for real-time actuation via Pololu Maestro controllers.

Engineering Physics Propulsion Lab (EPPL) Member

Embry-Riddle Aeronautical University

Jan 2024 – Dec 2025 Daytona Beach, FL

- Supported autonomous UAV swarm data collection using CAD, embedded systems, Mission Planner, and ArduPilot.
- **OpenMutt — Open-Source Quadruped Robot**
  R&D Member

  Developed an open-source quadruped robot to support multidisciplinary research using ROS 1 Noetic and custom embedded control.
- **SUAVE — Swarm Unmanned Aerial Vehicles using Emergence**
  R&D Member

  Conducted research and development on UAV swarm mapping technologies through simulation, embedded systems integration, and mission control logic.

Mechanical Engineer - Intern

Bosch USA

Aug 2024 – Dec 2024 Charleston, SC

- Contributed to process modification and optimization through design, operations, and documentation projects supporting automotive component manufacturing and overall equipment effectiveness (OEE).

Intern

TargetArm

Jul 2018 – Sep 2022 Brookfield, CT

- Manufactured 3D-printed components, managed a small-scale prototype production line, and used CAD software and embedded systems for field testing. Assisted in developing a mobile autonomous UAV launch and recovery system using Rust.